Discrete r.v.s

$X \sim \mathrm{Bin}(n,p)$, $\quad p(x)=\binom{n}{x}p^x(1-p)^{n-x}$, $\quad x=0,1,\dots,n.$

$X \sim \mathrm{Poisson}(\lambda)$, $\lambda>0$, $\quad p(x)=\frac{\lambda^x}{x!}\cdot e^{-\lambda}$, $\quad e\approx 2.71828\ldots$

$np \approx \lambda \qquad np=\lambda_n \xrightarrow{n\to\infty} \lambda$

$$\binom{n}{x}p^x(1-p)^{n-x} = \frac{n(n-1)\cdots(n-x+1)}{x!}\,p^x(1-p)^n/(1-p)^x =$$

$$= \frac{(\overset{\nearrow\lambda}{np})(\overset{\nearrow\lambda}{np}-p)\cdots(np-(x-1)p)}{x!}\cdot\left(1-\frac{\lambda_n}{n}\right)^n\cdot\frac{1}{(1-p)^x}\xrightarrow{n\to\infty}\frac{\lambda^x}{x!}\cdot e^{-\lambda}.$$

$$\left(1+\frac{a_n}{n}\right)^n \xrightarrow{n\to\infty} e^a \quad \text{if} \quad a_n\to a$$

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80 000 marriages (yearly)

$\mathbb{P}$(there will be a couple with the same birthday of the year)

$1-\left(1-\frac{1}{365}\right)^{80000}$

X: the number of couples with the stated property $\sim \mathrm{Bin}\left(80000, \frac{1}{365}\right)$

$\mathbb{P}(X\geq 1) = 1-\mathbb{P}(X=0) = \ldots$

Let Y be the number of couples who have their birthday on Nov. 21.

$\mathbb{P}(Y\geq 1) = 1-\mathbb{P}(Y=0) = 1-\left(1-\frac{1}{365^2}\right)^{80000}$

Using Poisson:

$\mathbb{P}(X \geq 1)$ $\qquad X \sim Bin\left(86000, \frac{1}{365}\right)$

$$\lambda = n \cdot p = \frac{86000}{365}$$

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0) = 1 - \frac{\lambda^0}{0!} \cdot e^{-\lambda} = 1 - e^{-\lambda}$$

Geometric r.v.

$\mathbb{P}(A) = p > 0$

X is the number of the first experiment when A happens, $X \in \{1, 2, \cdots\}$

$$p(x) = (1-p)^{x-1} \cdot p, \quad x = 1, 2, \cdots$$

$$\sum p(x) = 1 \; ; \quad \sum_{x=1}^{\infty} (1-p)^{x-1} \cdot p = 1$$

HyperGeometric r.v.

Box, N balls, M red, $N-M$ white, we draw $\underline{n}$ balls (without replacement)

X: the number of red balls drawn

$$p(x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, \cdots, n$$

$$\sum p(x) = 1 \rightsquigarrow \sum_{x=0}^{n} \binom{M}{x}\binom{N-M}{n-x} = \binom{N}{n}.$$

$NegBin(n, p) \sim X \longrightarrow$ Ross

Expected value of a r.v. X

$\mathbb{E}X$

$$x_1, x_2, \dots, x_n \qquad p(x_i) = p_i$$
$$p_1, p_2, \dots, p_n$$

N times play the game

$$\frac{p_1 N x_1 + p_2 N x_2 + \dots + p_n N x_n}{N} \approx \text{average win}$$

$$\mathbb{E}X = p_1 x_1 + \dots + p_n x_n$$

die $\qquad X \in \{1, 2, \dots, 6\}$

$$\mathbb{E}X = \frac{1}{6}\cdot 1 + \dots + \frac{1}{6}\cdot 6 = 3.5$$

$X_1 + X_2 \qquad X_1, X_2$ the numbers rolled with a die.

x_1+x_2	2	3	…	12
p	$\frac{1}{36}$	$\frac{2}{36}$		$\frac{1}{36}$

$$\mathbb{E}(X_1 + X_2) = 7$$

$$\mathbb{E}(X_1 + X_2) = \mathbb{E}X_1 + \mathbb{E}X_2$$

$$\mathbb{E}(cX) = c\,\mathbb{E}X$$

$$\mathbb{E}c = c$$

$$\sum g(x)\,p(x) = \mathbb{E}g(X)$$

$$\mathbb{E}X = \sum_{p(x)>0} x \cdot p(x)$$

4 school buses with 35, 38, 42, 46 students

X: is the number of students in the bus of a random student

Y: is the number of students in the bus of driver.

$\mathbb{E}X > \mathbb{E}Y$.